

Protein design

Part 2: predicting mutation free energy

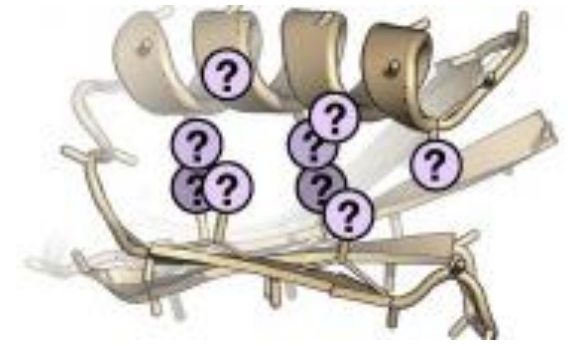
Anastassia Vorobieva



Introduction to protein design

Understanding amino acid patterns, predicting the effect of mutations

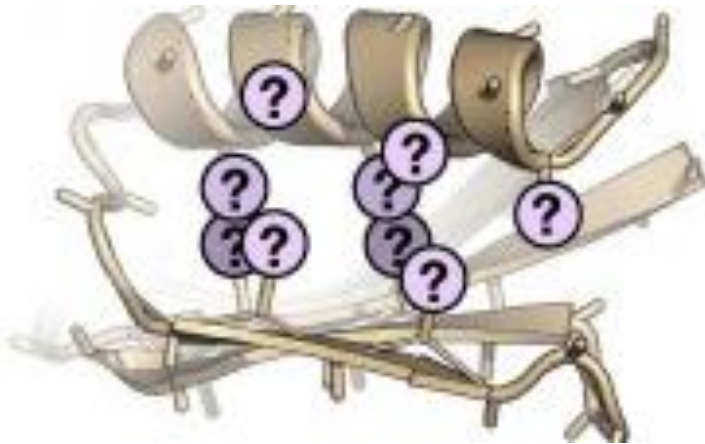
Protein design



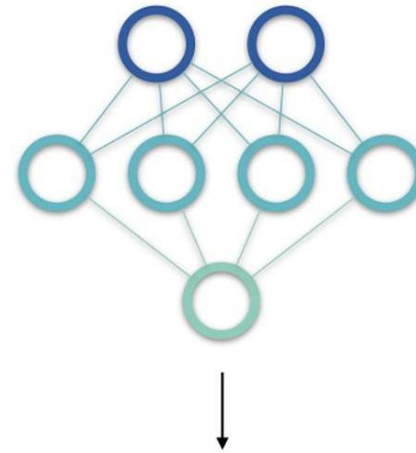
- **Protein design** is the process of manipulating a protein sequence to achieve a specific functions, structures, or stability.

Structure vs sequence-based protein design

Structure-based design



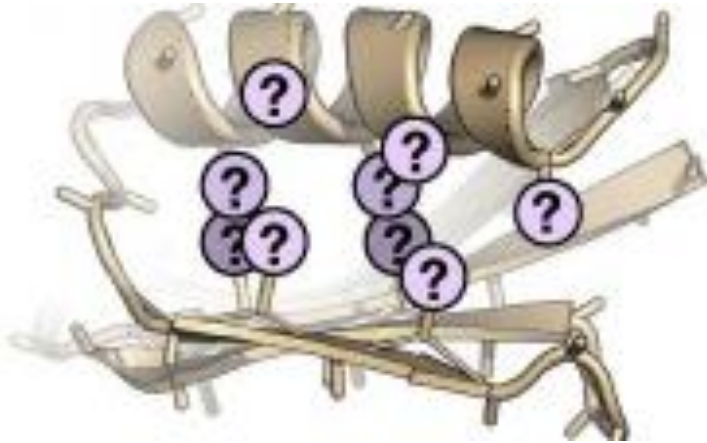
Direct sequence generation (e.g. LLMs)



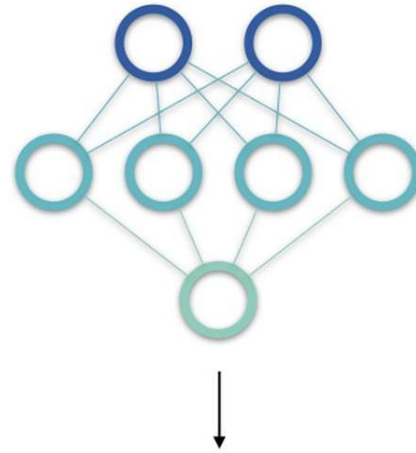
...MALKIPTHNHM...
...VFRDCEWS...
...WYIOPMNVGTDEW...

Structure vs sequence-based protein design

Structure-based design

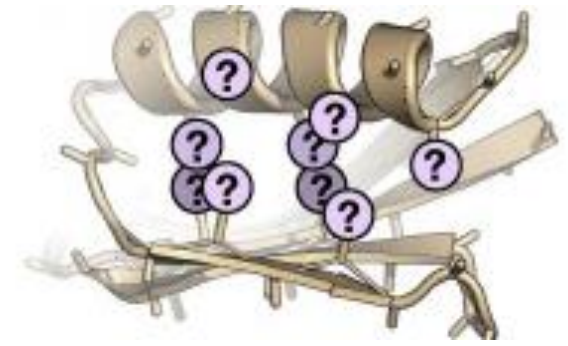


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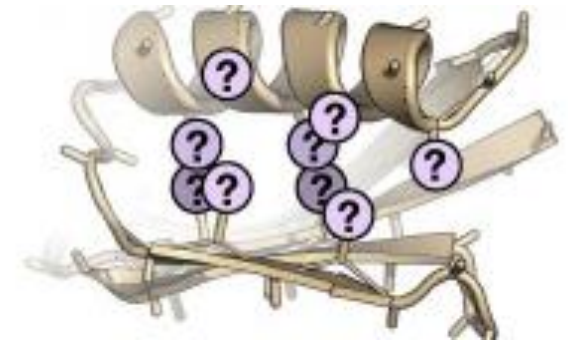
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Protein design



- **Protein design** is the process of manipulating a protein sequence to achieve a specific functions, structures, or stability.
- **Applications** of protein design:
 - Biomedicine: design binders, improve enzyme stability for gene therapy
 - Biotechnology: improve the stability and function of enzymes for cleaning, chemical synthesis, bio-remediation
 - Synthetic biology

Protein design



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- **Applications** of protein design:
 - Biomedicine: design binders, improve enzyme stability for gene therapy
 - Biotechnology: improve the stability and function of enzymes for cleaning, chemical synthesis, bio-remediation
 - Synthetic biology
- Predicting the Effect of Mutations on Protein Stability and Function
Understanding mutation effects can help in the design of proteins that are more stable, more active, or more specific to their targets

KumaMax: Designed Gluten Busting Enzyme Therapy for Celiac Disease

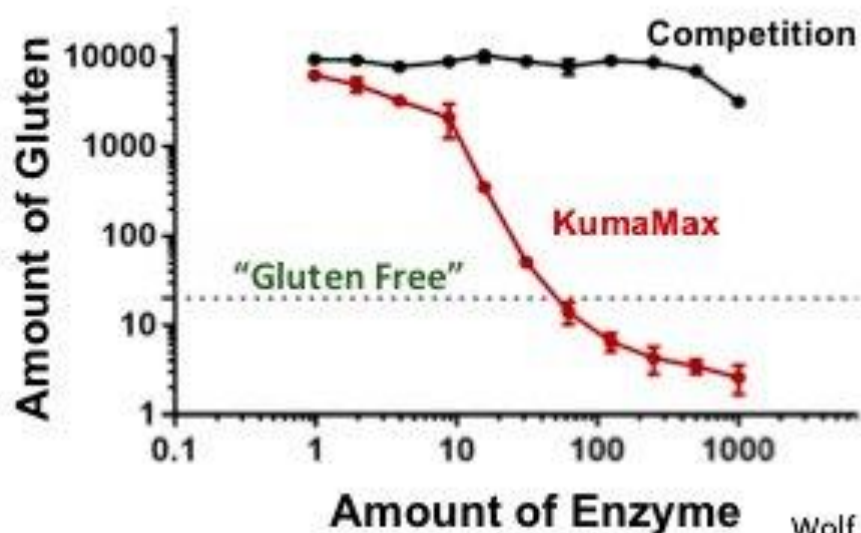
- Computationally re-designed enzyme
- Specific for gluten, destroys the immunogenic peptide
- Active only in low pH of the stomach
- Rapid and near complete degradation of gluten in beer, bread, etc.



Oral Pill

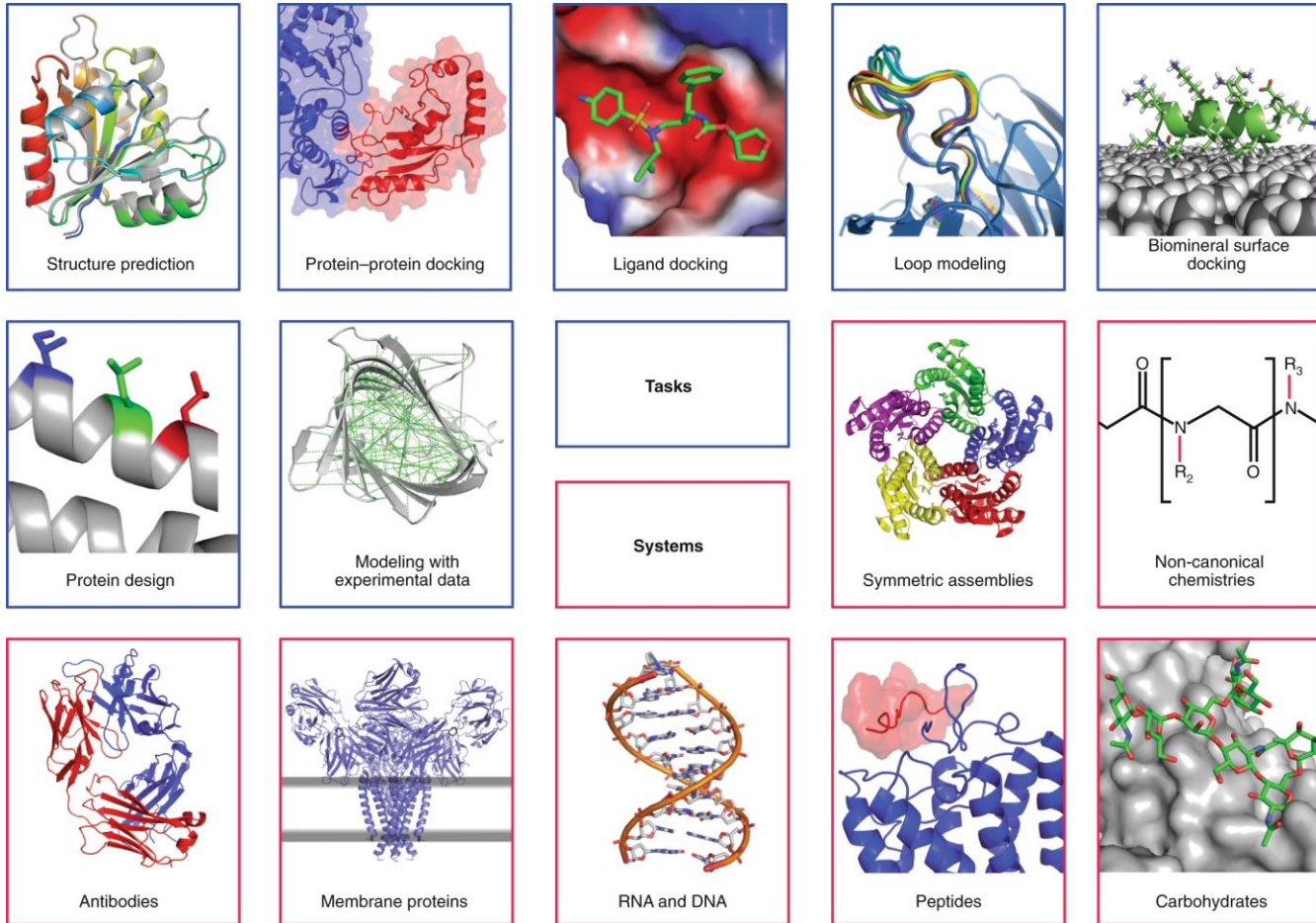


KumaMax
Designed Enzyme



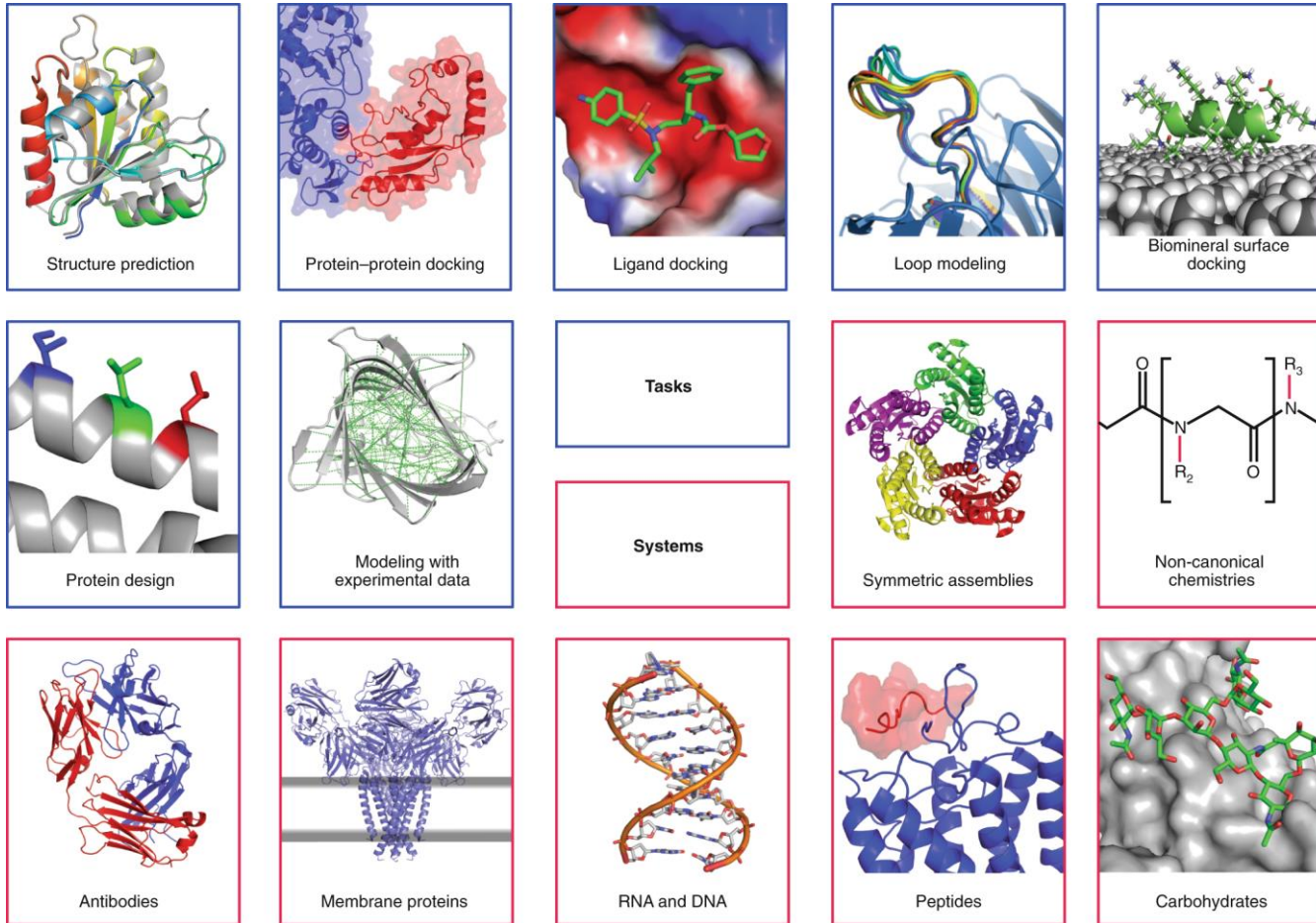
B. Preparing the PDB for Rosetta calculations

What is Rosetta ?

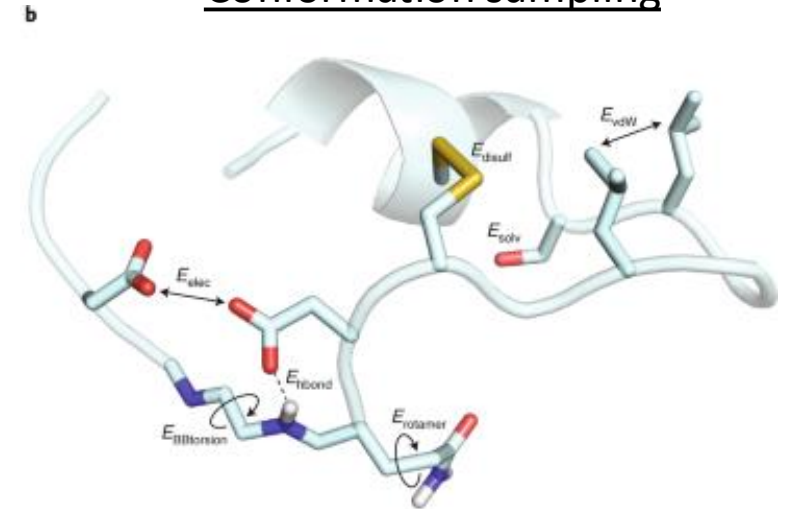


B. Preparing the PDB for Rosetta calculations

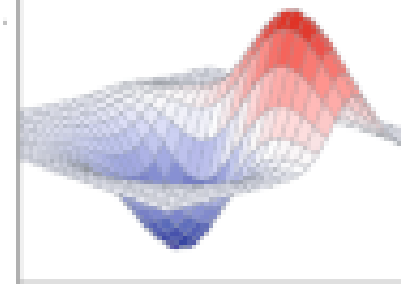
What is Rosetta ?



Conformation sampling



ScoreFunction



Scoring

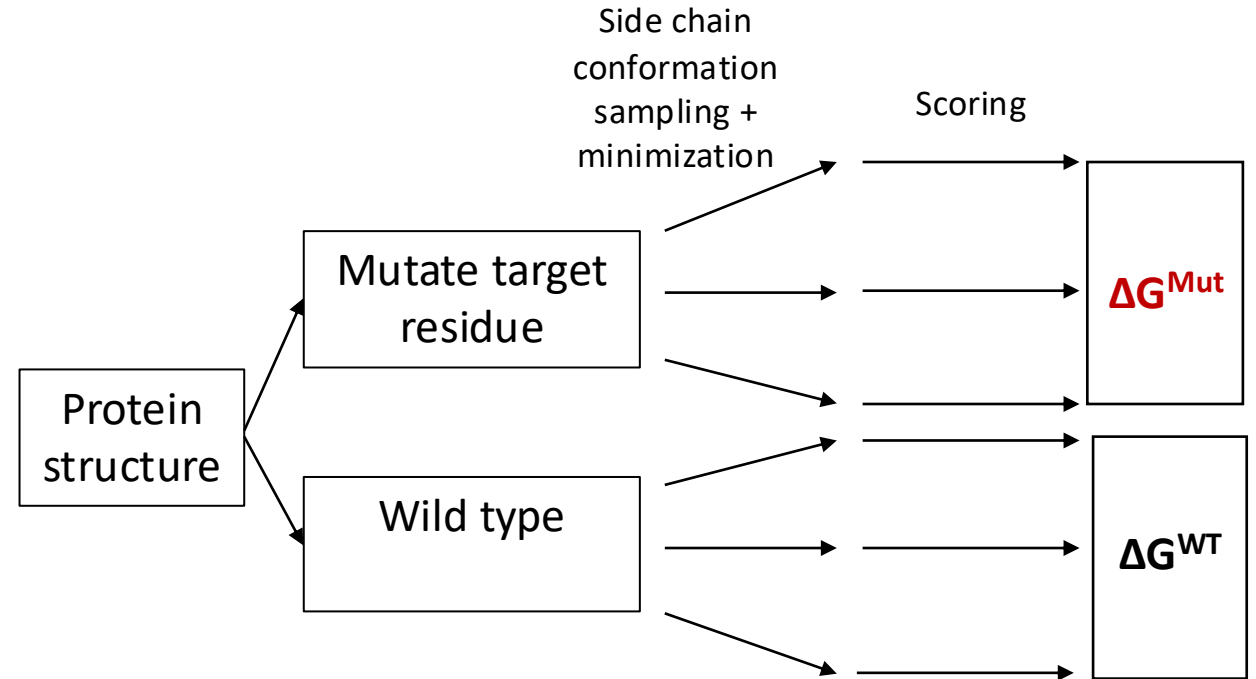
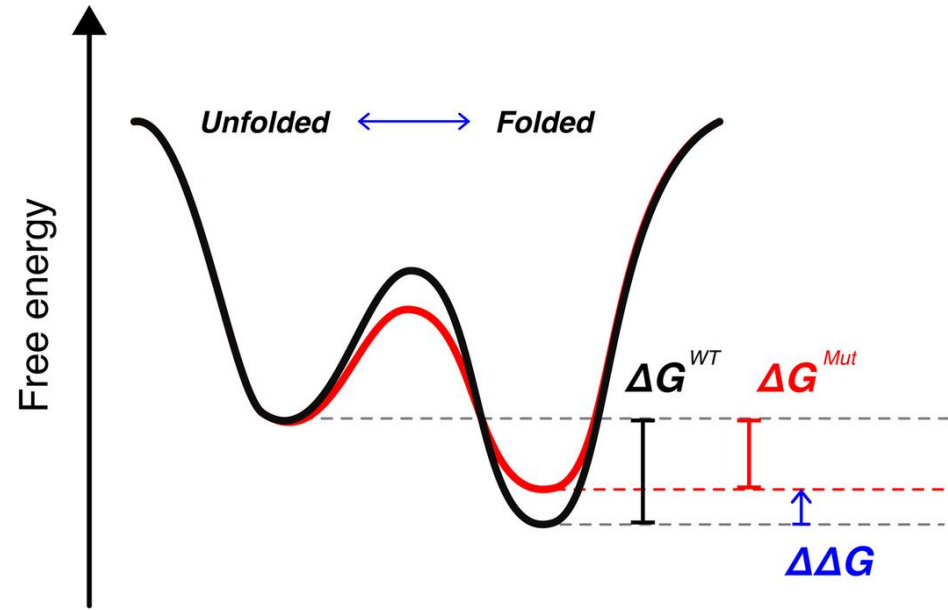
E_{vdw} Lennard-Jones for attractive or repulsive interaction
 E_{bond} Hydrogen bonding allows buried polar atoms
 E_{elec} Electrostatic interaction between charges
 E_{disulf} Disulfide bonds between cysteines

E_{solv} Implicit solvation model penalizes buried polar atoms
 $E_{torsion}$ Backbone torsion preferences from main-chain potential
 $E_{rotamer}$ Side-chain torsion angles from rotamer library
 E_{ref} Unfolded state reference energy for design

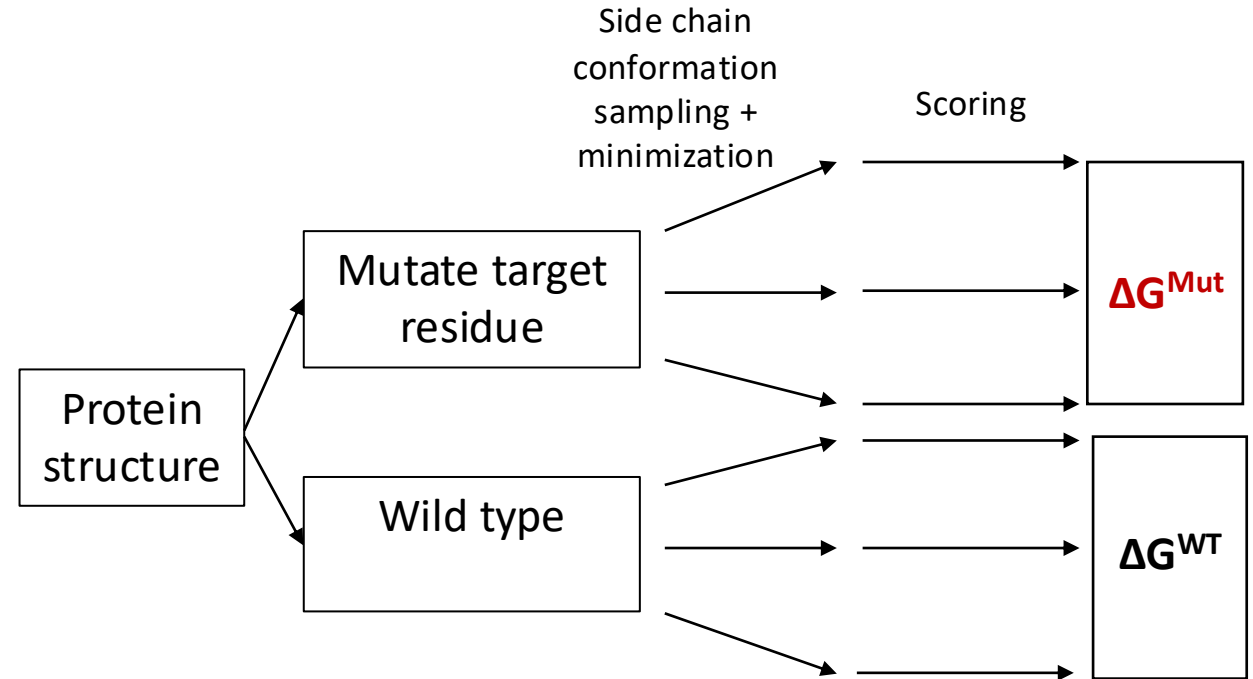
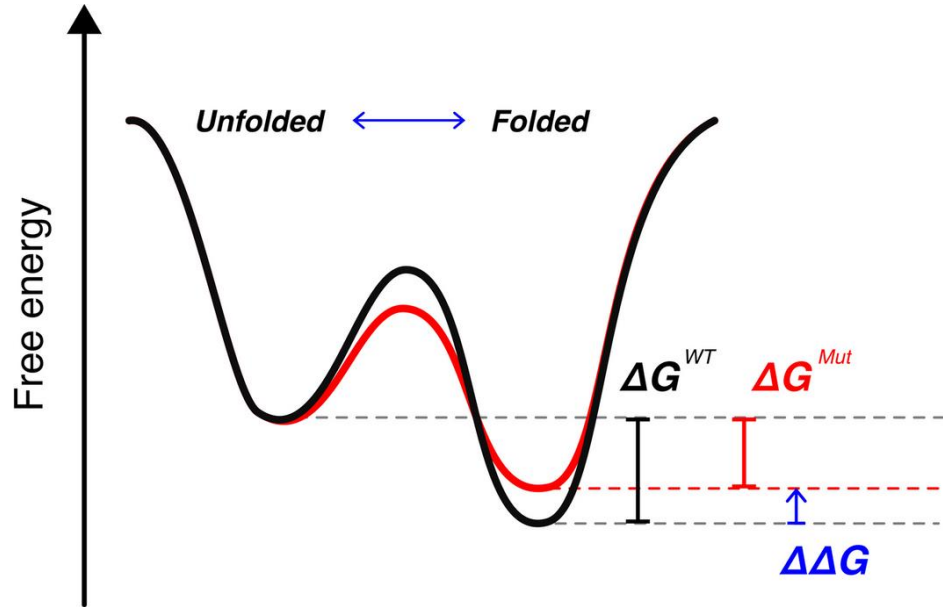
The PDB file is a list of atom coordinates

ATOM	8178	CD2	LEU	A	207	135.703	156.448	175.146	1.00	59.65		C
ATOM	8179	N	LEU	A	208	138.542	160.482	176.163	1.00	64.39		N
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HETATM	8195	C6	LMT	E	301	135.530	136.801	155.206	1.00	20.00		C
HETATM	8196	C7	LMT	E	301	136.343	137.543	156.242	1.00	20.00		C
HETATM	8197	C8	LMT	E	301	135.478	138.063	157.375	1.00	20.00		C
HETATM	8198	C9	LMT	E	301	136.194	137.846	158.690	1.00	20.00		C
HETATM	8199	C10	LMT	E	301	135.627	138.700	159.797	1.00	20.00		C
HETATM	8200	C11	LMT	E	301	135.963	138.052	161.121	1.00	20.00		C
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Running DDG calculations – the principle



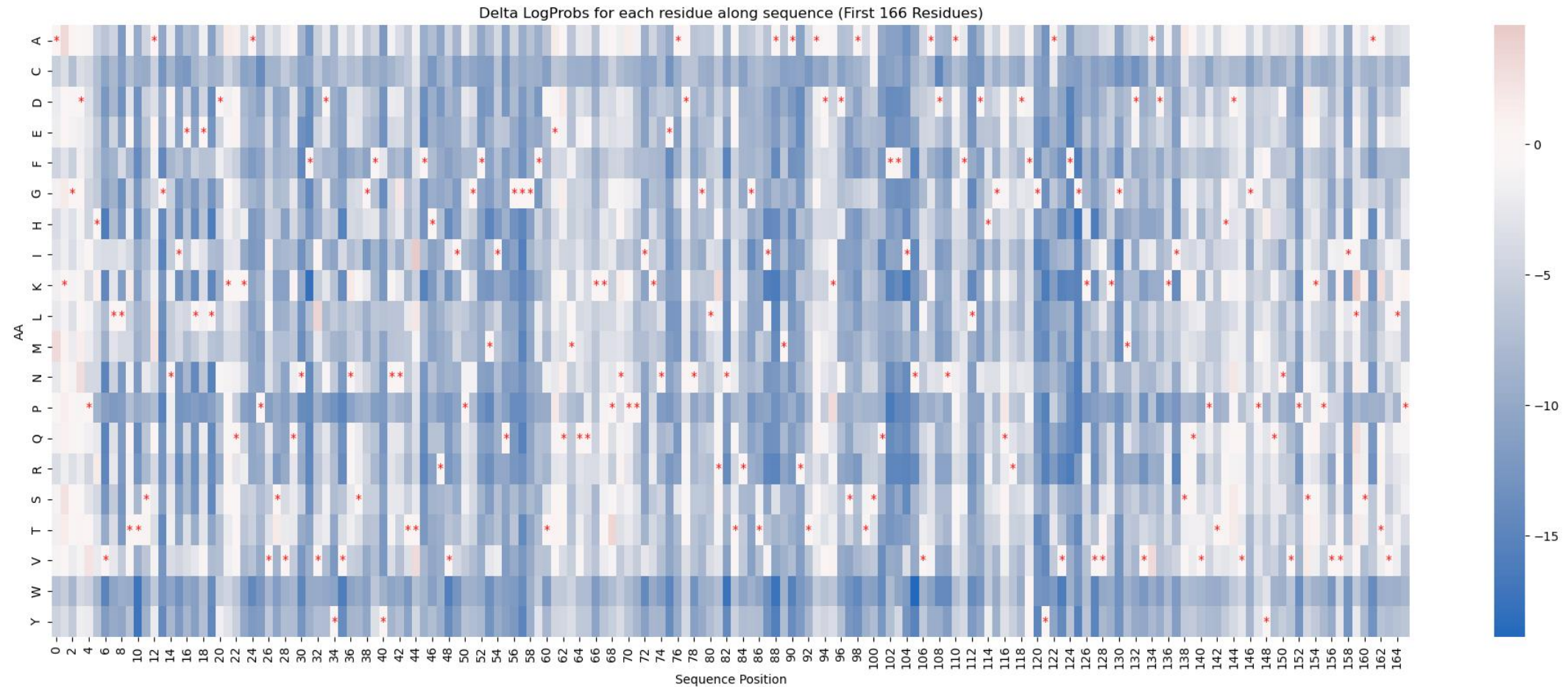
Running DDG calculations – the principle



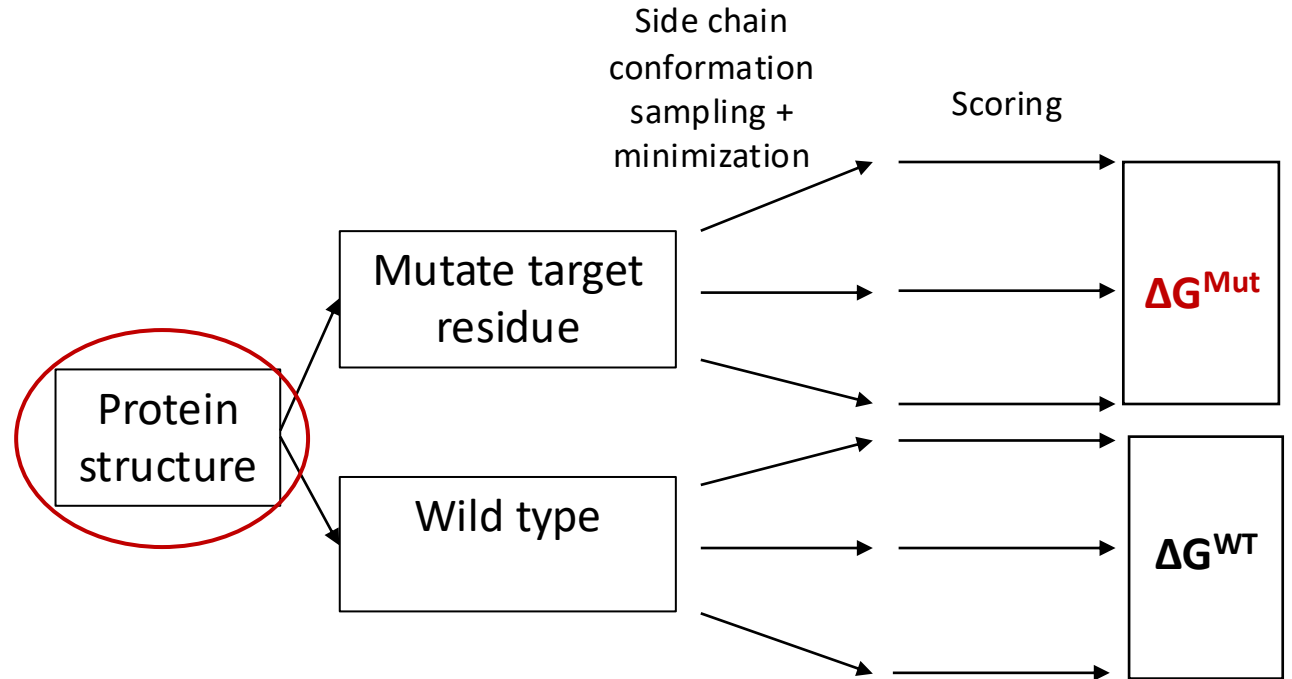
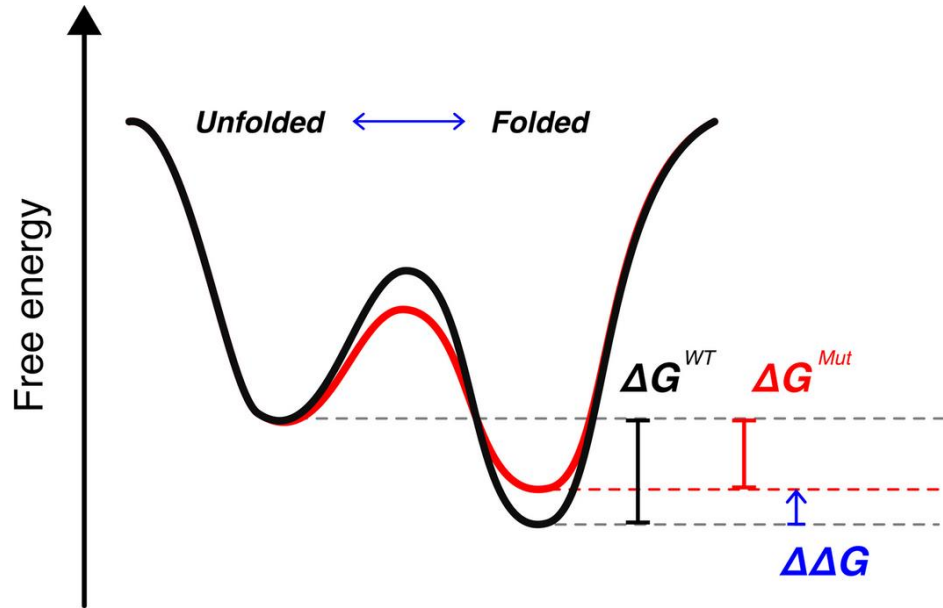
$$\Delta\Delta G = \frac{\sum_i MUT_total_score_i}{ddg_iterations} - \frac{\sum_i WT_total_score_i}{ddg_iterations}$$

Tools: FoldX, Rosetta

Running DDG calculations – the **heatmap** output



Running DDG calculations – the principle



$$\Delta\Delta G = \frac{\sum_i MUT_total_score_i}{ddg_iterations} - \frac{\sum_i WT_total_score_i}{ddg_iterations}$$

Tools: FoldX, Rosetta

1. Selecting your template PDB structure

Biological Assembly 1 ?



7FAO

Top7 surface mutant K42A Q43A K46A K57S K58S, and I68R

PDB DOI: <https://doi.org/10.2210/pdb7FAO/pdb>

Classification: **DE NOVO PROTEIN**

Organism(s): [unidentified](#)

Expression System: [Escherichia coli BL21](#)

Mutation(s): No ?

Deposited: 2021-07-07 Released: 2022-05-18

Deposition Author(s): [Ito, Y.](#), [Makabe, K.](#)

Experimental Data Snapshot

Method: X-RAY DIFFRACTION

Resolution: 1.43 Å

R-Value Free: 0.242

R-Value Work: 0.203

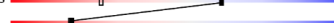
R-Value Observed: 0.205

Starting Model: experimental

[View more details](#)

wwPDB Validation ?

3D Report Full Report

Metric	Percentile Ranks	Value
Rfree		0.242
Clashscore		2
Ramachandran outliers		0
Sidechain outliers		1.8%
RSRZ outliers		6.8%

Worse Better

■ Percentile relative to all X-ray structures

□ Percentile relative to X-ray structures of similar resolution

Explore in 3D: [Structure](#) | [Sequence Annotations](#) | [Electron Density](#) | [Validation Report](#) | [Ligand Interaction \(MG\)](#)

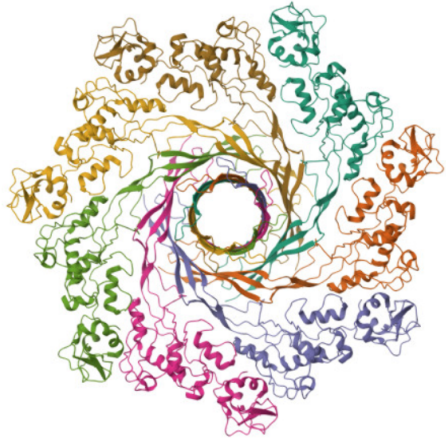
Global Symmetry: Asymmetric - C1 ?

Global Stoichiometry: Monomer - A1 ?

[Find Similar Assemblies](#)

Problem: Low resolution structure

Biological Assembly 1 ?
folding_assay folder for storing your documents.



Explore in 3D: [Structure](#) | [Sequence Annotations](#) | [Electron Density](#) | [Validation Report](#) | [Predict Membrane](#) ⓘ

Global Symmetry: Cyclic - C7 ⓘ ([Explore in 3D](#))
Global Stoichiometry: Homo 7-mer - A7 ⓘ

[Find Similar Assemblies](#)

5JZT

Cryo-EM structure of aerolysin pore in LMNG micelle

PDB DOI: <https://doi.org/10.2210/pdb5JZT/pdb> EM Map EMD-8187: [EMDB EMDDataResource](#)

Classification: **TOXIN**

Organism(s): [Aeromonas hydrophila](#)

Expression System: [Escherichia coli BL21\(DE3\)](#)

Mutation(s): No ⓘ

Membrane Protein: Yes ⓘ [OPM](#) [PDBTM](#)

Deposited: 2016-05-17 Released: 2016-07-13

Deposition Author(s): [Iacovache, I.](#), [Zuber, B.](#)

Funding Organization(s): Swiss National Science Foundation

Experimental Data Snapshot

Method: ELECTRON MICROSCOPY

Resolution: 7.40 Å

Aggregation State: PARTICLE

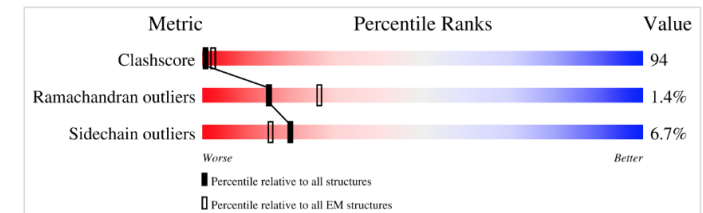
Reconstruction Method: SINGLE PARTICLE

Starting Model: experimental

[View more details](#)

wwPDB Validation ⓘ

[3D Report](#) [Full Report](#)



Problem: Poor refinement

Biological Assembly 1 ?



Explore in 3D: [Structure](#) | [Sequence Annotations](#) | [Electron Density](#) | [Validation Report](#) | [Ligand Interaction \(GOL\)](#) | [Predict Membrane](#) ⓘ

Global Symmetry: Asymmetric - C1 ⓘ
Global Stoichiometry: Monomer - A1 ⓘ

[Find Similar Assemblies](#)

2F1V

Outer membrane protein OmpW

PDB DOI: <https://doi.org/10.2210/pdb2F1V/pdb>

Classification: **MEMBRANE PROTEIN** TMB12_3_nb3

Organism(s): *Escherichia coli* K-12

Expression System: *Escherichia coli*

Mutation(s): No ⓘ

Membrane Protein: Yes ⓘ OPM PDBTM MemProtMD

Deposited: 2005-11-15 Released: 2006-01-24

Deposition Author(s): van den Berg, B.

Experimental Data Snapshot

Method: X-RAY DIFFRACTION

Resolution: 2.70 Å

R-Value Free: 0.314

R-Value Work: 0.292

R-Value Observed: 0.293

Starting Model: experimental

[View more details](#)

Display Files ▾

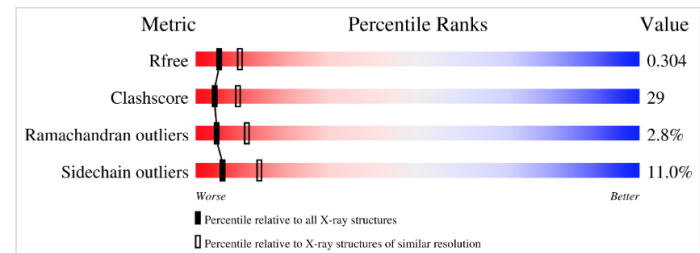
Download Files ▾

Data API

wwPDB Validation ⓘ

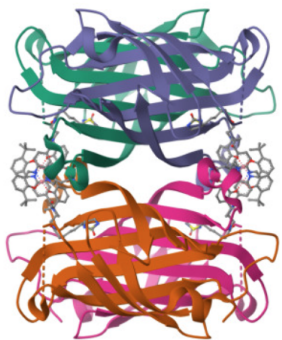
3D Report

Full Report



Problem: Missing loops, alternative conformations (b-factor)

Biological Assembly 1 ⓘ



Explore in 3D: [Structure](#) | [Sequence Annotations](#) | [Validation Report](#) | [Ligand Interaction \(UFU\)](#)

Global Symmetry: Dihedral - D2 ⓘ ([Explore in 3D](#))
Global Stoichiometry: Homo 4-mer - A4 ⓘ

[Find Similar Assemblies](#)

Biological assembly 1 assigned by authors and generated by PISA (software)

Biological Assembly Evidence: gel filtration

Macromolecule Content

8QEX

Streptavidin variant with a cobalt catalyst for CH metal-catalyzed hydrogen-atom-transfer (M-HAT)

PDB DOI: <https://doi.org/10.2210/pdb8QEX/pdb>

Classification: METAL BINDING PROTEIN
Organism(s): *Streptomyces avidinii*
Expression System: *Escherichia coli* BL21(DE3)
Mutation(s): No ⓘ

Deposited: 2023-09-01 **Released:** 2024-07-31
Deposition Author(s): Jakob, R.P., Chen, D., Ward, T.R.
Funding Organization(s): Not funded

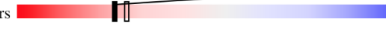
Experimental Data Snapshot

Method: X-RAY DIFFRACTION
Resolution: 1.90 Å
R-Value Free: 0.254
R-Value Work: 0.229
R-Value Observed: 0.230
Starting Model: experimental
[View more details](#)

wwPDB Validation ⓘ

3D Report

Full Report

Metric	Percentile Ranks	Value
Rfree		0.258
Clashscore		9
Ramachandran outliers		0
Sidechain outliers		0
RSRZ outliers		5.7%

Worse

Better

■ Percentile relative to all X-ray structures
▨ Percentile relative to X-ray structures of similar resolution

Ligand Structure Quality Assessment ⓘ

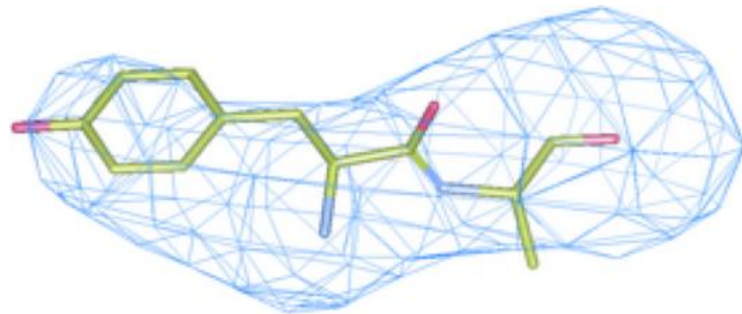
Worse 0



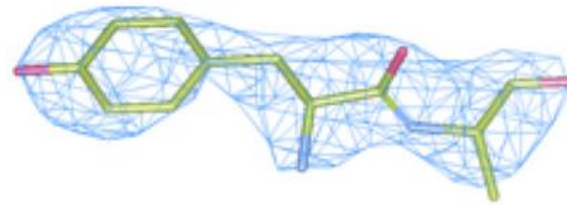
1 Better

Ligand structure goodness of fit to experimental data

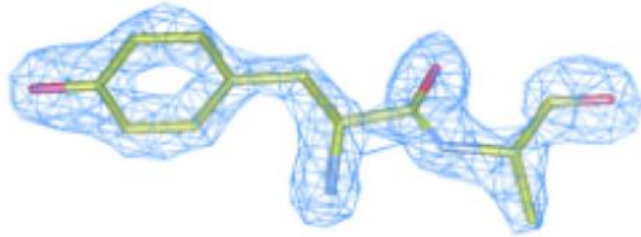
Problem: Ligand modelling!!



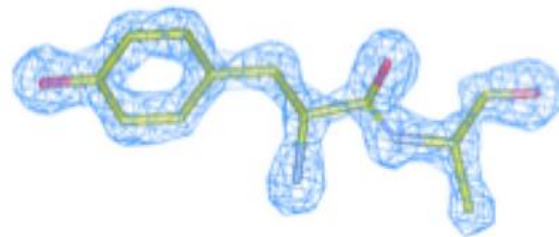
5.0 Å resolution, 2.5σ



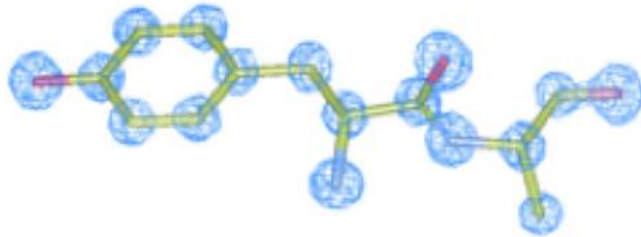
3.0 Å resolution, 6.5σ



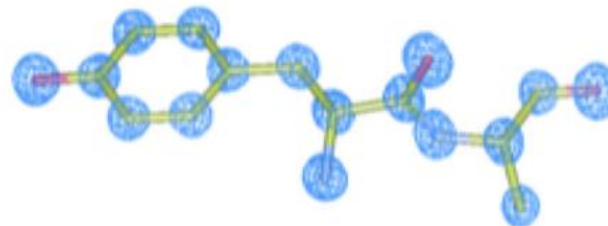
2.0 Å resolution, 8.5σ



1.5 Å resolution, 10σ



1.0 Å resolution, 14σ



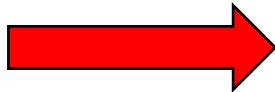
0.6 Å resolution, 12σ

Q: What are the different steps to prepare the protein structure for energy calculations?

A. The first step is to clean the PDB (remove non-protein atoms)

A. Clean PDB

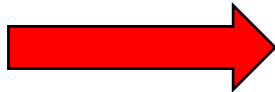
ATOM	8178	CD2	LEU	A	207	135.703	156.448	175.146	1.00	59.65		C
ATOM	8179	N	LEU	A	208	138.542	160.482	176.163	1.00	64.39		N
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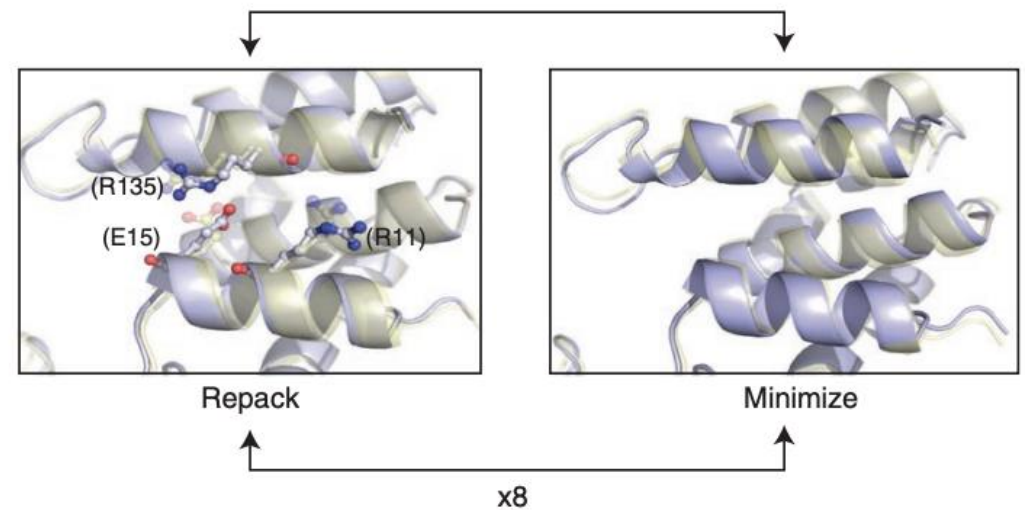
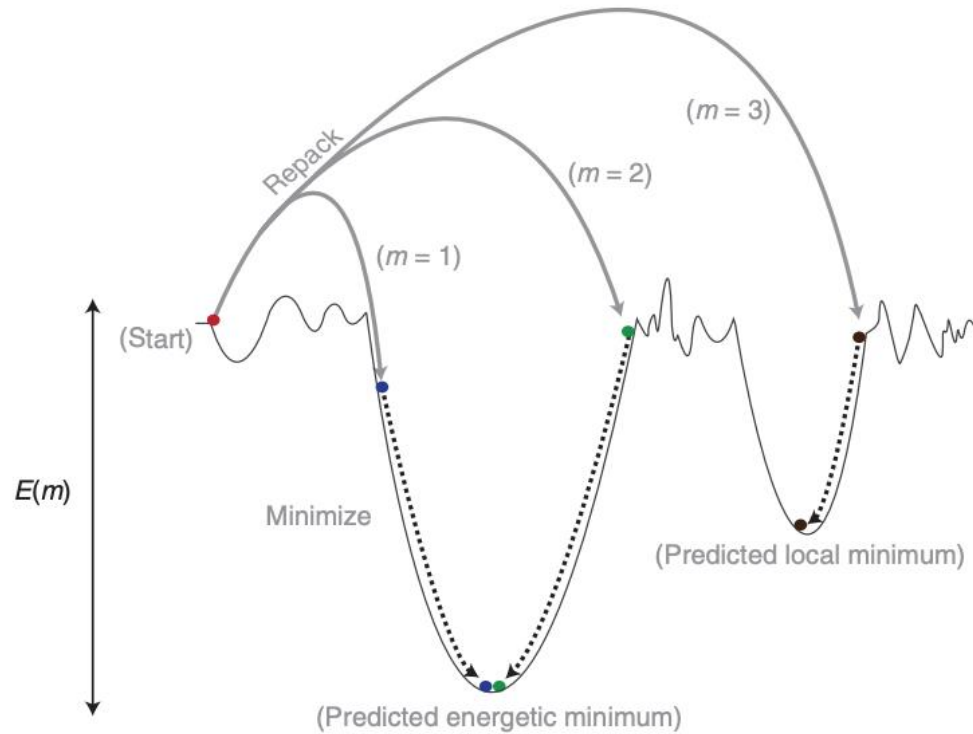
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HETATM	8193	C4	LMT	E	301	136.376	135.025	153.714	1.00	20.00		C
HETATM	8194	C5	LMT	E	301	135.935	135.347	155.125	1.00	20.00		C
HETATM	8195	C6	LMT	E	301	135.530	136.801	155.206	1.00	20.00		C
HETATM	8196	C7	LMT	E	301	136.343	137.543	156.242	1.00	20.00		C
HETATM	8197	C8	LMT	E	301	135.478	138.063	157.375	1.00	20.00		C
HETATM	8198	C9	LMT	E	301	136.194	137.846	158.690	1.00	20.00		C
HETATM	8199	C10	LMT	E	301	135.627	138.700	159.797	1.00	20.00		C
HETATM	8200	C11	LMT	E	301	135.963	138.052	161.121	1.00	20.00		C
HETATM	8201	C12	LMT	E	301	135.929	139.084	162.215	1.00	20.00		C



B. Relaxing the PDB for Rosetta calculations

Rosetta FastRelax algorithm
~ simulated annealing

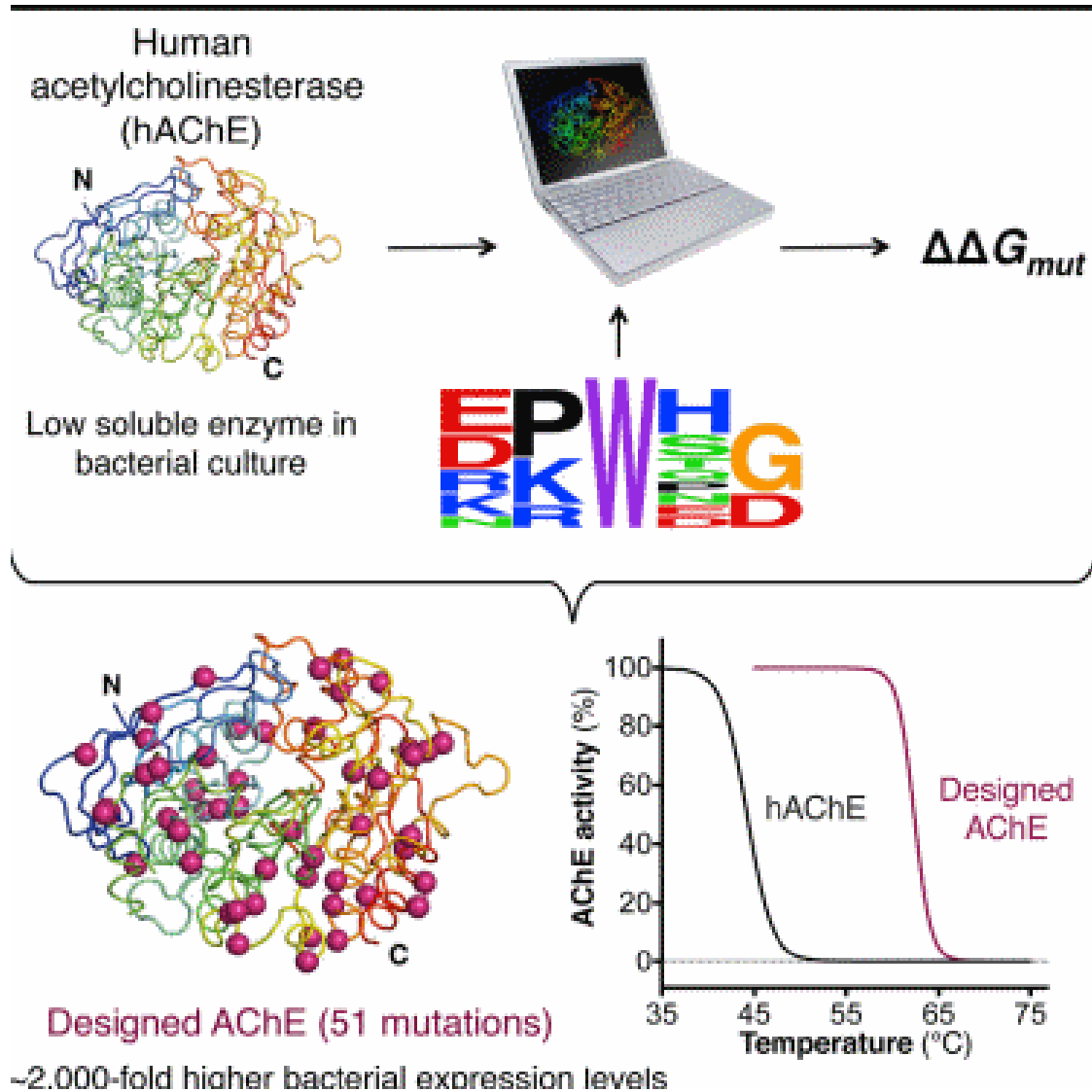


Q: Calculate the effect of mutations on protein-protein interactions?

Q: Calculate the effect of mutations on protein-protein interactions?

$$\Delta\Delta G_{interface} = \frac{\sum_{i=0}^n COMPLEX_score_n}{n} - \frac{\sum_{i=0}^n SEPARARATE_score_n}{n}$$

Running Combinatorial design - PROSS



Improves bacterial expression, stability and function of many eukaryotic (human) proteins

Paper: <https://doi.org/10.1016/j.molcel.2018.08.033>

Server: <https://pross.weizmann.ac.il/>

Next lecture: *de novo* protein design